An Isometric Operator-Algebra Action Which Is Not Completely Isometric

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1 Question

Wang and Zhu ask in Remark 2.4(i) of arXiv:2305.03933 whether their p-nuclearity theorem for crossed products remains true when a p-completely isometric action is replaced by an isometric action. They also state the subsidiary question whether an isometric action is necessarily p-completely isometric. The latter has a negative answer as written, already when $p = 2$.

2 Construction

Let e_{ij} denote the standard matrix units in M_3, acting on $H = \ell_2^3$. Put

$$C_1 = e_{21}, \qquad C_2 = e_{31}, \qquad R_1 = e_{12} = C_1^{\mathsf{t}}, \qquad R_2 = e_{13} = C_2^{\mathsf{t}},$$

and define

$$B_C = \operatorname{span}\{I, C_1, C_2\}, \qquad B_R = \operatorname{span}\{I, R_1, R_2\}.$$

All products $C_j C_k$ and $R_j R_k$ vanish. Hence B_C and B_R are unital commutative operator algebras. Transpose gives a complex-linear algebra isomorphism

$$\theta : B_C \longrightarrow B_R, \qquad \theta(T) = T^{\mathsf{t}}.$$

For the Hilbert-space operator norm, $\|T^{\mathsf{t}}\| = \|T\|$; equivalently, T and T^{t} have the same singular values. Thus θ is an isometry at level one.

Represent

$$A = B_C \oplus B_R \subset B(H \oplus_2 H)$$

as a block-diagonal unital operator algebra, and define

$$\alpha(b, c) = \big(\theta^{-1}(c), \theta(b)\big).$$

The direct-sum norm is the maximum of the two component norms. Therefore α is an isometric algebra automorphism and $\alpha^2 = \mathrm{id}$. It generates an isometric action of the finite amenable group $\mathbb{Z}_2$.

Proposition 1. *The action generated by α is not 2-completely isometric.*

Proof. In $M_2(A)$ consider

$$X = \begin{bmatrix} (C_1, 0) & 0 \\ (C_2, 0) & 0 \end{bmatrix}.$$

The nonzero direct-sum component of X is the operator

$$\mathsf{C} = \begin{bmatrix} C_1 & 0 \\ C_2 & 0 \end{bmatrix} \quad \text{on } H \oplus_2 H.$$

If $\xi = (\xi_1, \xi_2)$ and $\xi_1 = (z_0, z_1, z_2)$, then

$$\mathsf{C}\xi = (C_1\xi_1, C_2\xi_1), \qquad \|\mathsf{C}\xi\|^2 = 2|z_0|^2.$$

Consequently $\|X\| = \|\mathsf{C}\| = \sqrt{2}$, with equality attained at $\xi_1 = (1, 0, 0)$ and $\xi_2 = 0$.

Entrywise application of α gives

$$\alpha^{(2)}(X) = \begin{bmatrix} (0, R_1) & 0 \\ (0, R_2) & 0 \end{bmatrix}.$$

Its nonzero component is

$$\mathsf{R} = \begin{bmatrix} R_1 & 0 \\ R_2 & 0 \end{bmatrix}.$$

For the same vector ξ,

$$\mathsf{R}\xi = (R_1\xi_1, R_2\xi_1), \qquad \|\mathsf{R}\xi\|^2 = |z_1|^2 + |z_2|^2.$$

Thus $\|\alpha^{(2)}(X)\| = \|\mathsf{R}\| = 1$. Since $\sqrt{2} \neq 1$, the second amplification of α is not isometric. □

3 Scope

The example is finite-dimensional, unital, and nondegenerately represented, and its acting group is finite. It therefore gives a full negative answer to the auxiliary automatic-complete-isometry question at $p = 2$.

It does not decide whether the crossed product in the example is 2-nuclear exactly when A is 2-nuclear. Nor does this row/column proof extend verbatim to $p \neq 2$: the level-one column and row norms then become the ℓ_p and ℓ_q norms and are no longer equal. Hence the main relaxed theorem and its intended non-Hilbertian range remain outside the claim of this packet.

4 Reference

Z. Wang and S. Zhu, *p-nuclearity of L^p-operator crossed products*, arXiv:2305.03933.